

Healthcare/IoT Cybersecurity Testbed: Sem. 2 Milestone 4

By: Justin Bower, Nathan Maloney, and Ipule Pipi

Instructors/Advisors: Dr. Sudhakaran and Dr. Aydeger

Testbed user

Local testbed — client machine

Device connection

USB / wireless capture
Adafruit BLE sniffer
EEPROM & chip readers

RE & analysis tools

Ghidra
Wireshark / Tshark
Binwalk & Foremost

Exploitation scripts

GDB / pwndbg (ret2*)
Pre-made & custom scripts

AWS cloud — server side

Web dashboard

React + Tailwind UI
Cognito authentication

API + compute

API Gateway
Lambda: instant + durable

Storage & sandbox

DynamoDB + S3 storage
Docker sandboxed analysis
CVE database

Uploads firmware, results & CVE data

System Architecture Explained

Main groups:

1. Prospective Users: (Security Researchers, Manufacturers, Engineers, and Advisors)
2. User-facing systems: Web Dashboard (Server-side) and Local GUI (Client-machine)
3. Internal Systems:
 - AWS Cloud backend (includes DynamoDB, S3 Storage, Backend API, and VM Compute)
 - Standard commercial router flashed with OpenWRT and utilizing Tailscale or base Wireguard for tunnelling

Any internet access for testing will be provided by a dedicated ethernet line granted to us in the IoT lab.

Milestone 4 Progress Summary

Task	% Complete	Justin	Nathan	Ipule	Jordan	To-Do
Project Management, Doc Creation, and Planning	100%	100%	0%	0%	0%	Continue managing project and creating documents as planning is now complete
Complete Firmware and Adafruit Demos	100%	50%	25%	25%	0%	Web interface demo of database functions
VR/RE & CVE database work	75%	50%	0%	0%	25%	Finish VR/RE of smart watches/heart monitors and enter any CVES
Local GUI development	75%	25%	0%	0%	50%	Add launch options, create more launch scripts, make GUIs launch in-frame

Milestone 4 Progress Summary

Testbed Set-up	100%	34%	0%	33%	33%	Flash the router with OpenWRT
Web Dashboard & Cloud Backend Integration for	0%	0%	60%	40%	0%	Integrate analysis initiation, job status, results/artifacts, workflow states, and authorization into dashboard
Cloud infrastructure & backend services	90%	0%	90%	0%	0%	Cloud API endpoints and firewall for local application
Artifact & Firmware Management	100%	0%	50%	50%	0%	Continue integration with analysis workflow

Testbed Review & Progress



- This is still the current deployment of the physical testbed, featuring the virtual components, as follows:
 - Server-side -> Web Dashboard (Left Laptop)
 - Local-Side -> Analysis Tooling (Right Laptop)
- Router needs to be flashed with OpenWRT and a VPN tunnel with [Insert Chosen Application will be required.
- We will be integrating more devices as the RE/VR advances
- So far we have reverse engineered the oximeters and are now moving onto smart watches/heart monitors

Research Publication

Dr. Sudhakaran will be including us in a paper to publish our research.

This is the PhD paper she asked us to study and use for planning our own sensor research.

PREPRINT: SUBMITTED TO IEEE PAINÉ 2026

1

SentryBus: A Multi-Vantage Observability Model and Validated Instrument for I²C Sensor-Interface Manipulation

Sandesh More*, Elton Batista*, Karla Daley*, Sneha Sudhakaran*

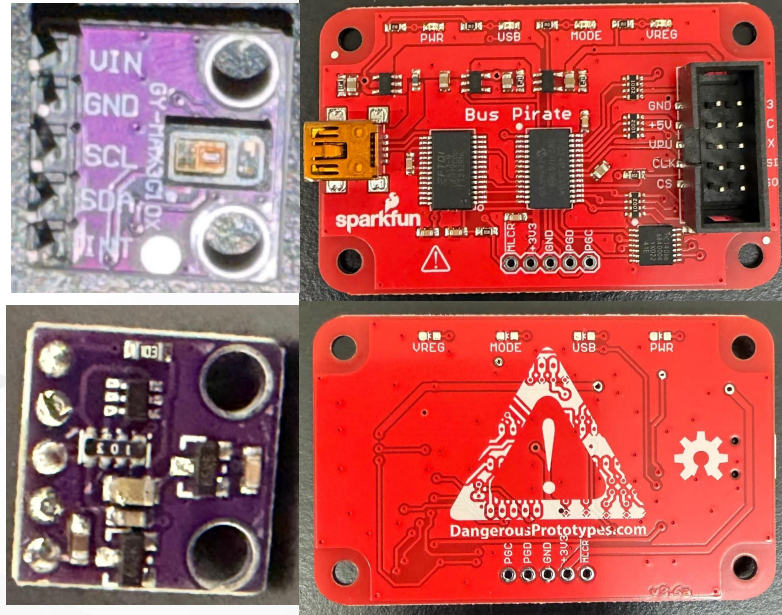
*Department of Computer Engineering and Sciences
Florida Institute of Technology, Melbourne, FL, USA

I2C Sensor Research and Bus Pirate

To the right is the DS18B20 temperature sensor that we will research.

This sensor is usually found in oximeters, general vitals monitors, and other diagnostic medical devices.

This will be our first experience with sensor research.



Main Dashboard and Device List

Here is the status of all items for reverse engineering.

Once they have all been reverse engineered, and known CVEs entered in the database page, then we can begin doing our own vulnerability research.

Then, we can start implementing scripts, and automating parts of the vulnerability research process.

Device List

Devices			15 total	
DEVICE	ROLE	STATUS OF BEING REVERSED		
Alfa Networks AWUS036ACH	owner	Complete	View Firmware	Upload Firmware
Fitbit Charge 6	owner	In-Progress	View Firmware	Upload Firmware
ENOMIR GTX12 Smartwatch BK NEW	owner	In-Progress	View Firmware	Upload Firmware
Alfa Network AWUS036ACS	owner	Complete	View Firmware	Upload Firmware
Garmin HRM-DUAL	owner	In-Progress	View Firmware	Upload Firmware
Wearpulse Oxyfit PO6	owner	In-Progress	View Firmware	Upload Firmware
Beurer PO30	owner	Complete	View Firmware	Upload Firmware
Beurer Elite 950	owner	In-Progress	View Firmware	Upload Firmware
EMAY Pulse Oximeter EMO-80	owner	Complete	View Firmware	Upload Firmware
Innovo iP900BP-B	owner	Complete	View Firmware	Upload Firmware
Beurer PO30	owner	Complete	View Firmware	Upload Firmware
Alfa Network AWUS036ACM	owner	Complete	View Firmware	Upload Firmware
Beurer PO60	owner	Complete	View Firmware	Upload Firmware
Panda NWireless AC600	owner	Complete	View Firmware	Upload Firmware
Polar H10	owner	In-Progress	View Firmware	Upload Firmware

Artifacts Upload

This is status of the Artifacts page upon uploading a file.

For now, we created a report of CVE entries
Once, the database is live then we can associate them with specific devices.

Artifacts

Files from your reverse engineering work, ready for analysis

Upload files

Firmware, pcaps, logs, binaries or anything else, up to 5 GiB each. Files uploaded together form one upload batch.

TAGS (ALL FILES)

lab-2, capture-day-1

DEVICES (ALL FILES, OPTIONAL)

Alfa Networks AWUS036ACH

Fitbit Charge 6

ENOMIR GTX12 Smartwatch BK NEW

Alfa Network AWUS036ACS

Garmin HRM-DUAL

Wearpulse Oxyfit PO6

Beurer PO30

Beurer Elite 950

EMAY Pulse Oximeter EMO-80

Innovo IP900BP-B

Beurer PO30

Alfa Network AWUS036ACM

Beurer PO60

Panda NWireless AC600

Polar H10

1 of 1 ready

128.4 KiB / 128.4 KiB

FILE	SIZE	TYPE	VERSION	STATUS
wifi_chipset_cve_report.pdf	128.4 KiB	other	—	Ready

New upload

Your artifacts

Upload batch 01a0fd98 x

Run a script on this upload

All types

Filter by tag

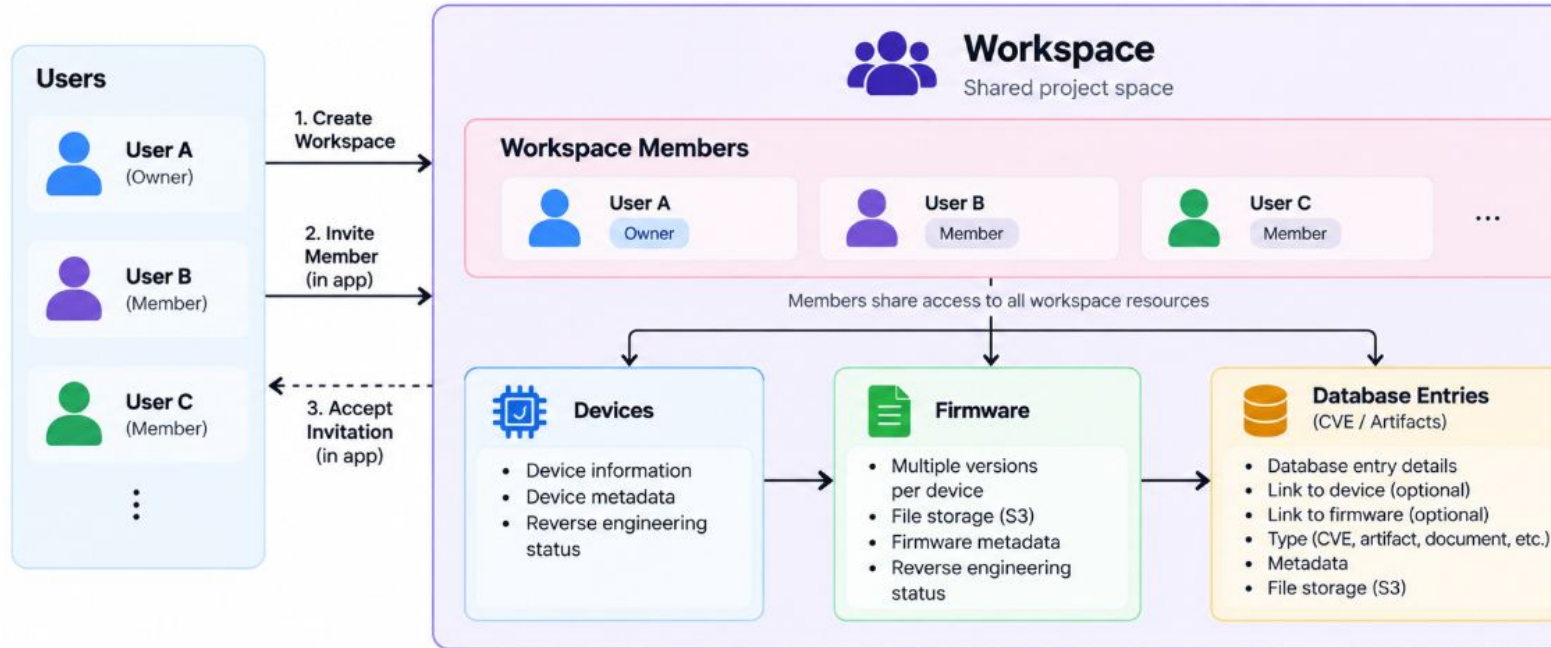
Refresh

☐	NAME	TYPE	STATUS	SIZE	TAGS / DEVICES	UPLOADED		
☐	wifi_chipset_cve_report.pdf	other	ready	128.4 KiB	Alfa Networks AWUS036ACH Alfa Network AWUS036ACM	Alfa Network AWUS036ACS Panda NWireless AC600	10/2/2026, 1:10:03 PM	Download

Database (Implementation)

Workspace Architecture

A shared project space where multiple users collaborate on devices, firmware, and database entries.



CVE Report -> Database Entries

 **Vzoniq**
Healthcare IoT Vulnerability T...

RESEARCH

- Dashboard
- Devices
- Firmware
- Artifacts
- Vulnerabilities**

ANALYSIS

- Scripts
- Environments
- Runs

Workspaces

Usage

Research / Vulnerabilities

Healthcare IoT Test...MembersIP

Vulnerabilities (CVE)

Known vulnerabilities recorded against your devices · Workspace · Healthcare IoT Testbed

All severitiesAll devices

Refresh

CVE ID	SEVERITY	CVSS	AFFECTED CHIPSETS	DEVICES	UPDATED	ACTIONS
CVE-2025-8302 Heap overflow vulnerability in the Windows driver / rtl81xx SDK fa...	High	—	RTL8811AU	—	Oct 6, 2026	
CVE-2025-8300 Heap overflow vulnerability in the Windows driver / rtl81xx SDK fa...	High	—	RTL8811AU	—	Oct 6, 2026	
CVE-2025-8299 Heap overflow vulnerability in the Windows driver / rtl81xx SDK fa...	High	—	RTL8811AU	—	Oct 6, 2026	
CVE-2025-8298 Out-of-bounds read in N6CQueryInformationHandleCustomized11...	Low	—	RTL8811AU	—	Oct 6, 2026	
CVE-2025-8301 Heap buffer overflow in N6CSet DOT11 CIPHER DEFAULT KEY (rt...	High	7.8	RTL8811AU	—	Oct 6, 2026	
CVE-2020-26652 Denial-of-service vulnerability in nl80211 send chandef affecting t...	High	7.5	RTL8812AU	—	Oct 6, 2026	
CVE-2022-26441	Medium	6.7	MT7610UN MT7612UN	—	Oct 6, 2026	

Live Demo of Script Runs

Server Dashboard URL: <https://vzoniq.com>

Server Feature Requirements

- The testbed tightly integrates many disparate tools, automates firmware extraction, firmware analysis, entropy analysis, decryption, into one platform, is a novel approach to cybersecurity analysis.
- The testbed reduces the background knowledge needed for manufacturers or researchers to reverse-engineer Medical/IoT devices while unifying the reverse-engineering toolset.
- The testbed is specifically tailored for IoT and medical hardware, which is often proprietary and notoriously difficult to analyze.
- The testbed allows for direct firmware extraction from connected devices via standard interfaces and therefore streamlines the process of obtaining firmware/binaries for vulnerability assessment.
- The testbed allows for built-in entropy analysis combined with pre-existing decryption techniques to uncover encrypted or obfuscated firmware sections more efficiently.
- The testbed has support for both ready-made exploitation scripts and user-developed custom scripts provides flexibility to confirm known weaknesses or explore new attack vectors.

Server Feature Evaluation

Below is our evaluation for the server features:

- The testbed tightly integrates many disparate tools, automates firmware extraction, firmware analysis, entropy analysis, decryption, into one platform, is a novel approach to cybersecurity analysis. ✗
- The testbed reduces the background knowledge needed for manufacturers or researchers to reverse-engineer Medical/IoT devices while unifying the reverse-engineering toolset. ✓ 100%
- The testbed is specifically tailored for IoT and medical hardware, which is often proprietary and notoriously difficult to analyze. ✓ 100%
- The testbed allows for direct firmware extraction from connected devices via standard interfaces and therefore streamlines the process of obtaining firmware/binaries for vulnerability assessment. ✓ 100%
- The testbed allows for built-in entropy analysis combined with pre-existing decryption techniques to uncover encrypted or obfuscated firmware sections more efficiently. ✓ 50%
- The testbed has support for both ready-made exploitation scripts and user-developed custom scripts provides flexibility to confirm known weaknesses or explore new attack vectors. ✓ 100%

We will list additional tests, especially for security, to ensure we are providing the maximum functionality.

User Testing Requirements

- Users can connect a variety of IoT and medical devices that use standard interfaces, wired or wireless, to the testbed, enabling full device access, interaction, and manipulation.
- Users can extract firmware directly from connected IoT and medical devices, obtaining complete firmware/binaries for detailed examination and vulnerability assessment.
- Users can analyze firmware entropy to evaluate randomness and compression characteristics
- Users can utilize industry-standard open-source tools to perform reverse-engineering tasks and comprehensive vulnerability research.
- Users can input their own CVE lists, and automatically retrieve lists of known CVEs, which are stored server-side in a database.
- Users can utilize pre-made scripts or upload custom exploitation scripts to rapidly test known , suspected, or new vulnerabilities, and potential attack vectors.
- Users can perform all connection, extraction, analysis, and exploitation tasks through an intuitive client-side graphical interface, thereby streamlining the process for both novice and experienced users.
- Users can create and upload documentation, or integrate existing documentation via GitHub, with all materials stored in server-side storage.

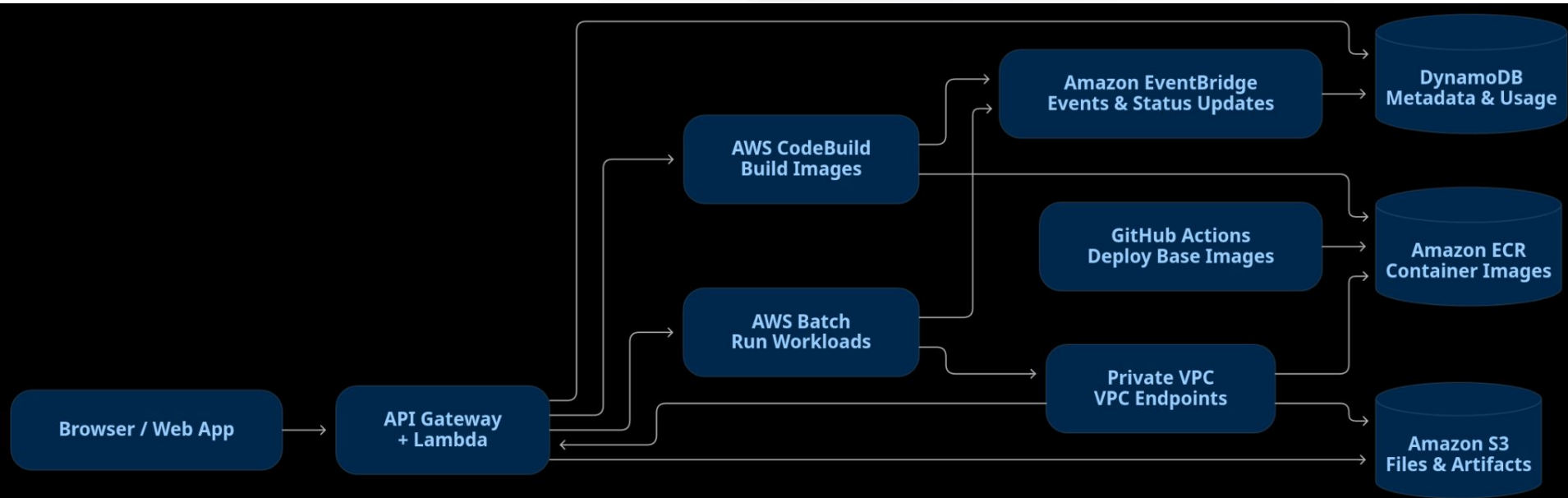
User Testing Evaluation

Below is our evaluation for the user tasks:

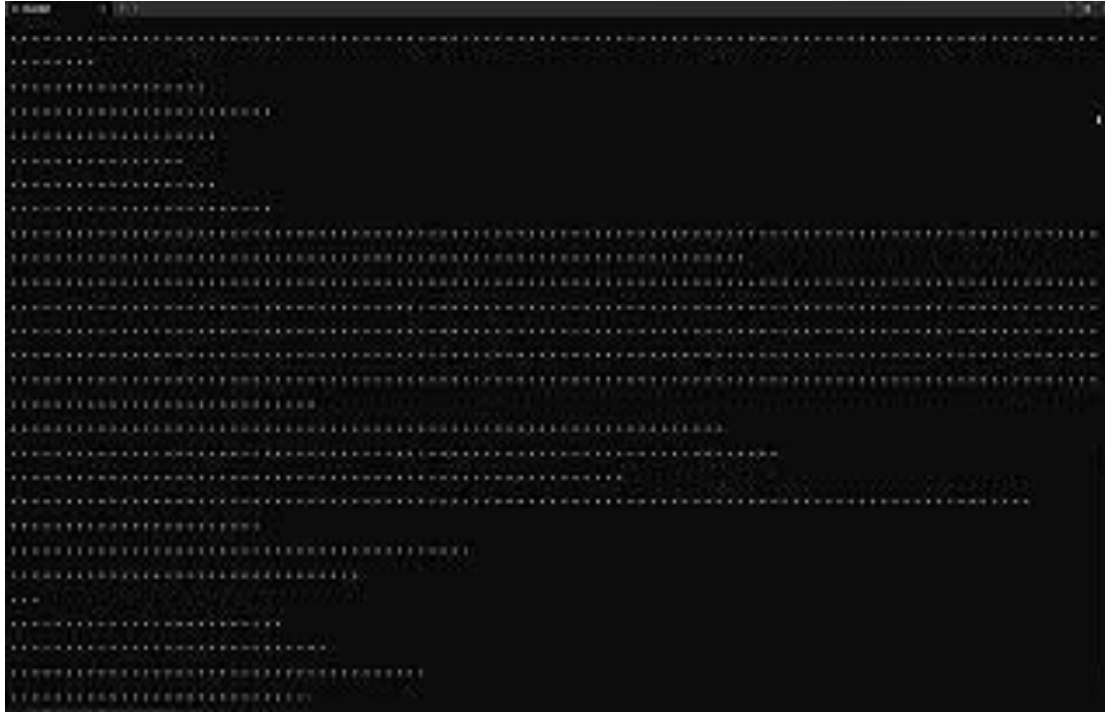
- Users can connect a variety of IoT and medical devices that use standard interfaces, wired or wireless, to the testbed, enabling full device access, interaction, and manipulation. ✓ 100%
- Users can extract firmware directly from connected IoT and medical devices, obtaining complete firmware/binaries for detailed examination and vulnerability assessment. ✓ 100%
- Users can analyze firmware entropy to evaluate randomness and compression characteristics ✓ 100%
- Users can utilize industry-standard open-source tools to perform reverse-engineering tasks and comprehensive vulnerability research. ✗
- Users can input their own CVE lists, and automatically retrieve lists of known CVEs, which are stored server-side in a database.
- Users can utilize pre-made scripts or upload custom exploitation scripts to rapidly test known , suspected, or new vulnerabilities, and potential attack vectors. ✓ 100%
- Users can perform all connection, extraction, analysis, and exploitation tasks through an intuitive client-side graphical interface, thereby streamlining the process for both novice and experienced users. ✓ 50%
- Users can create and upload documentation, or integrate existing documentation via GitHub, with all materials stored in server-side storage. ✓ 100%

We will add additional tests, especially for security, to ensure we are providing the maximum functionality.

New Scripting Architecture



Adafruit Bluetooth Sniffer Demo



Directory: C:\Users\Justin\AppData\Roaming\Nordic Semiconductor\Sniffer\logs

Mode	LastWriteTime	Length	Name
-a---	4/20/2026 7:54 PM	480130	lift_mouse_capture.pcap
-a---	4/20/2026 7:54 PM	572909	lift_mouse_log.txt

This preliminary demo shows step 1 of the whole workflow for Bluetooth RE

Wireshark interface showing a capture of Bluetooth RE workflow data. The capture is on interface `phy0.mon`, band `2.4 GHz`, channel `1 - 2.412 GHz`, and width `20 MHz`.

No.	Time	Source	Destination	Protocol	Length	Info
19	0.902644			Packet	49	
20	0.943649			Packet	49	
21	0.976620			Packet	49	
22	0.978067			Packet	62	
23	0.979766			Packet	62	
24	0.981221			Packet	58	
25	0.982445			Packet	58	
26	0.983448			Packet	58	
27	0.984219			Packet	38	
28	0.985240			Packet	63	
29	0.986395			Packet	58	
30	0.987427			Packet	63	
31	1.309093			Packet	63	
32	1.411937			Packet	63	
33	1.413488			Packet	58	
34	1.414989			Packet	58	
35	1.417220			Packet	58	

Frame 35: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits)

- Encapsulation type: USER 10 (55)
- Arrival Time: Apr 20, 2026 19:52:51.134095000 EDT
- UTC Arrival Time: Apr 20, 2026 23:52:51.134095000 UTC
- Epoch Arrival Time: 1776729171.134095000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 2.231000 milliseconds]
- [Time delta from previous displayed frame: 2.231000 milliseconds]
- [Time since reference or first frame: 1.417220000 seconds]
- Frame Number: 35
- Frame Length: 58 bytes (464 bits)
- Capture Length: 58 bytes (464 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: user_dlt:data]
- Character encoding: ASCII (0)

User encapsulation not handled: DLT=157, check your Preferences->Protocols->DLT_USER

[Expert Info (Warning/Undecoded): User encapsulation not handled: DLT=157, check your Preferences->Protocols->DLT_USER]

... Data ...

This screenshot shows step 2 of a Bluetooth RE workflow.

This PCAP shows what the Adafruit sensor picks up when listening.

This frame captured information for a Logitech wireless mouse.

Local GUI Progress



Milestone 5 Task Matrix

Task	Justin	Nathan	Ipule	Jordan
DS18B20 I2C Sensor/Buspirate Research and Integration	Spearhead Bus Pirate and sensor research/integration	Assist with the research and integration	Assist with the research and integration	Assist with the research and integration
Containerized Deployments and Platform Security Testing	Work with Jordan to port network tools to the GUI as needed and write deployment scripts for their cloud VM setup.	Assist with the creation and deployment of various docker containers	Assist with the creation and deployment of various docker containers	Work with Justin to port network tools to the GUI as needed and write deployment scripts for their cloud VM setup.
Start-to-Finish Demo	Verify integrations and create the demo	Create pipelines to allow for files to be easily used in each step of RE work	Create pipelines to allow for files to be easily used in each step of RE work	Create a firmware analysis / exploitation pipeline for extracted firmware
Project Management, Communication, and Documents	Manage project, documents, and communications,	Assist with document creation	Assist with document creation	Assist with document creation



Network Traffic Analysis Integration (Wireshark/Tshark)	Create & integrate Wireshark / Tshark scripts with Local GUI	Assist with testing	Assist with testing	Create & integrate Wireshark / Tshark scripts with Local GUI
Firmware RE & Analysis (Ghidra workflows, entropy analysis, Binwalk/Foremost segmentation)	Create & integrate Ghidra analysis / file-carving scripts with Local GUI	Assist with testing	Assist with testing	Create & integrate Ghidra analysis / file-carving scripts with Local GUI
WiFi/Bluetooth Exploitation Script Development	Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation	Assist with research, implementation, and testing	Assist with research, implementation, and testing	Create & integrate WiFi/Bluetooth analysis scripts with Local GUI
Local GUI Development	Assist Jordan with research, implementation, and testing	Create API to allow local GUI to communicate with cloud securely	Assist with API	Design UI/UX & integrate tools into application
Real time file analysis on cloud platform	Test security with Nathan by ensuring proper file parsing with no code injection	Create systems for users to be able to run real time file analysis on the cloud	Create GUI that allows user to see the notebook and its results in real time	Test security with Nathan by ensuring proper file parsing with no code injection

Thanks for listening...
Any questions?